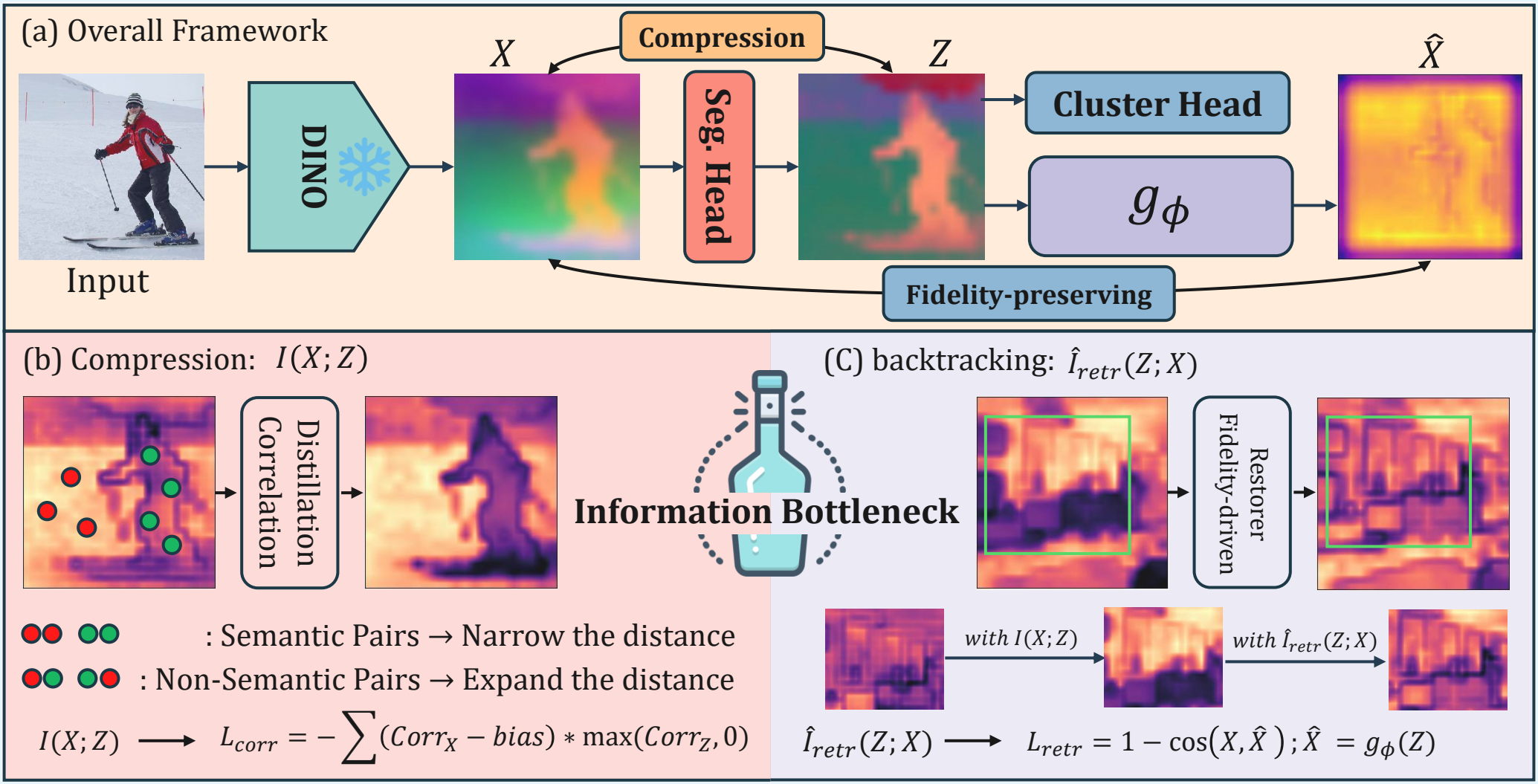
(a) Overall Framework
Input
DINO
X
Compression
Seg. Head
Z
Cluster Head
g_ϕ
$\hat{X}$
Fidelity-preserving
(b) Compression: $I(X;Z)$
Distillation
Correlation
Information Bottleneck
(C) backtracking: $\hat{I}_{retr}(Z;X)$
Restorer
Fidelity-driven
: Semantic Pairs → Narrow the distance
: Non-Semantic Pairs → Expand the distance
$I(X;Z) \longrightarrow L_{corr} = -\sum(Corr_X - bias) * \max(Corr_Z, 0)$
with $I(X;Z)$
with $\hat{I}_{retr}(Z;X)$
$\hat{I}_{retr}(Z;X) \longrightarrow L_{retr} = 1 - \cos(X, \hat{X}); \hat{X} = g_\phi(Z)$